

# Hari Vignesh Muralidharan

LinkedIn: [linkedin.com/in/harivigneshcse/](https://www.linkedin.com/in/harivigneshcse/)  
Github: [github.com/Hari-Vignesh214](https://github.com/Hari-Vignesh214)

Email: [harivignesheswari@gmail.com](mailto:harivignesheswari@gmail.com)  
Mobile: +1 (425) 520-1208  
Seattle, WA - Open to relocate

## SKILLS SUMMARY

- **Languages:** Python, C#, TypeScript, C++, JavaScript, Java, SQL, HTML5, CSS3
- **AI & Design:** LLM Orchestration, RAG, Agent-to-Agent (A2A), Multi-Agent Systems, GenAI Workflows  
Spec-Driven Development (SDD), Prompt Engineering
- **Frameworks:** .NET Core, ASP.NET, MVC.NET, Django, Angular, Spring, React.js, Node.js, Vue.js, Entity Framework, REST API, GraphQL, Flask, SOAP, Angular Material, Express.js, FastAPI, WPF, WCF, Web API
- **Libraries:** OpenCV, TensorFlow, PyTorch, NumPy, Pandas, Scikit-learn, ECharts, D3.js, LINQ, Bootstrap, jQuery
- **Cloud:** AWS(Amazon EC2, Amazon S3, Amazon SQS), Azure, GCP, Kafka, Kubernetes, Hadoop
- **Databases:** PostgreSQL, Dgraph, MongoDB, CosmosDB, Oracle PL/SQL, MS SQL Server, T-SQL
- **Security:** CSRF protection, XSS prevention, OAuth 2.0, JWT authentication, SAML, Role-based access control (RBAC)
- **Tools:** Swagger(OpenAPI), Git, TFS, Jenkins, Docker, Amazon CloudWatch, Apache, JUnit, JMP

## EDUCATION

- **University of Houston - Clear Lake** Houston, TX  
*Master of Science - Computer Science* May 2023  
*Courses:* Design and Analysis of Algorithms, Data Mining Tools and Techniques, Machine Learning, Database Systems
- **Anna University** Chennai, India  
*Bachelor of Engineering - Computer Science and Engineering* May 2018  
*Courses:* Design and Analysis of Algorithms, Compiler Design, Theory of Computation, Data Structures

## EXPERIENCE

- **Intel Corporation** Hillsboro, OR  
*Software Engineer* August 2023 - August 2025
  - **Automation & Reliability Engineering:** Automated data validation and reporting workflows while migrating a legacy JMP dashboard to **Angular Micro Frontends**, reducing manual analysis by **25+ hours/week**. Improved reliability using asynchronous API calls and **OnPush** change detection for datasets exceeding **1M+ records**.
  - **Data Engineering & Automation:** Designed and automated **petabyte-scale ETL pipelines** using Python and **Apache Airflow**, enabling reliable data ingestion from Hadoop to downstream services with **99.9% data integrity SLA** and reduced manual validation effort.
  - **AI/ML Integration:** Leveraged **AI frameworks** (scikit-learn, pandas, NumPy) to implement automated **anomaly detection, data quality checks, and ranking algorithms** on large-scale datasets processed using **Apache Spark**, enabling scalable data transformation pipelines for backend consumption.
  - **Full-Stack Development:** Built and maintained modular **SPAs** with **Angular Micro Frontends** and scalable APIs (**.NET Core, Django REST**), supporting **5000+** daily active users. Applied clean coding principles and design best practices, while partnering with cross-functional teams to deliver maintainable, reliable enterprise workflows.
- *Full-Stack Software Developer Intern* August 2022 - May 2023
  - **Legacy Migration:** Migrated a **graph analytics module** from a legacy **JMP dashboard** to a scalable web solution using **Angular, .NET Core, and Django**, improving accessibility and reducing load time by **40%**.
  - **Security & Data Integrity:** Engineered a secure access layer using **JWT** and **OAuth 2.0** to resolve data synchronization issues; implemented cache-invalidation logic that eliminated unauthorized access risks and ensured **100% real-time data accuracy**.
  - **Performance Optimization:** Implemented targeted **code refactoring** and **lazy loading** techniques, resulting in **35% improvement** in application performance and enhanced user experience.
- **Capgemini(GECAS, GE Aviation)** Bangalore, India  
*Senior Software Engineer* August 2018 - June 2021
  - **Technology Migration:** Led the successful migration of web application from **Classic ASP** to **ASP.NET MVC** framework, improving system performance by **45%** and reducing maintenance overhead by **30%**.
  - **Team Leadership:** Improved team performance by **20%** through thorough code reviews, code reusability initiatives, and implementation of **CSRF protection** to prevent unauthorized requests.
  - **Backend Development with Oracle PL/SQL:** Designed and optimized **Triggers, Procedures, and Functions** to automate workflows, enforce business rules at the database layer, and enhance **data integrity and system performance**, reducing manual interventions and ensuring consistency across the application.
  - **Client Delivery:** Consistently generated **160+ billable hours** monthly, saving clients **\$350-\$500 per month** through efficient development practices and automated solutions.

## PROJECTS

- **aiMarrvel Web App (MEAN [MongoDB, Express.js, Angular, Node.js], AWS, Jasmine, Artillery.io):** Developed a secure, web-based job submission system for the aiMarrvel tool (*in collaboration with Baylor College of Medicine, University of Houston-Clear Lake*), enabling users to upload VCF and HPO files to generate ranked variant outputs. Ensured data privacy by incorporating authentication and access controls in alignment with **HIPAA compliance** standards. Introduced batch processing for multi-file uploads and implemented a global variant search feature to filter rank matrix data efficiently. (2023)

## HONORS AND AWARDS

- Completed **Google Professional AI** Certification via Coursera, focusing on **Generative AI and ML** workflows.
- Led 50+ members as President of CSI Student Chapter, organizing 15+ technical events under Anna University.
- Presented award-winning research on "Smart Irrigation Based on IoT" at MIT and Sri Sairam College of Engineering.